# Smart Facility Issue Management Platform
Product Requirements Document (PRD)
Developed for
Medeniyet Technopark
Version 1.0

## 1. Project Overview

### 1.1 Introduction

TechFix is a modern web-based Facility Issue Management Platform developed specifically for Medeniyet Technopark.
The purpose of the platform is to provide a fast, simple and centralized way for companies operating inside the Technopark to report facility-related issues, technical faults, maintenance requests, missing equipment, complaints and suggestions.
Instead of relying on phone calls, emails or verbal communication, every issue is submitted digitally through a QR-code-based reporting system.
Administrators can then monitor, evaluate, prioritize and resolve every report through a centralized management dashboard.
The platform is designed to simplify communication, improve maintenance operations and create a fully digital issue management workflow.

### 1.2 Problem Statement

Currently, facility-related problems inside Technopark are generally reported through manual communication methods such as phone calls, messaging applications or direct communication with administrative staff.
These methods create several operational problems:
- Reports may be forgotten.
- Communication becomes inconsistent.
- There is no centralized archive.
- Multiple people may report the same issue.
- Administrators cannot easily track issue history.
- Companies cannot monitor the progress of their requests.
- There is no structured maintenance workflow.

As the number of companies grows, these problems become increasingly difficult to manage.

### 1.3 Proposed Solution

TechFix provides a centralized digital platform where companies can instantly report facility-related issues by simply scanning a QR code.
The system allows users to submit reports within less than one minute without creating an account.
Each submitted report is automatically stored in the system and becomes available for administrators to review.
Administrators can then classify, prioritize and manage every issue through an easy-to-use dashboard.
Every report remains traceable from creation until completion.

## 1.4 Project Objectives

The objectives of TechFix are:
• Digitalize facility issue reporting.
• Reduce communication delays.
• Simplify maintenance management.
• Improve response time.
• Eliminate paper-based processes.
• Reduce duplicate reports.
• Increase operational efficiency.
• Provide transparent issue tracking.
• Create a centralized issue archive.
• Improve overall user experience.

## 1.5 Scope

The first version of TechFix focuses on facility issue reporting and issue management inside Medeniyet Technopark.
The platform includes:
• QR-based issue reporting
• Administrator management panel
• Issue tracking
• Photo upload
• Report categorization
• Report status management
• Dashboard
• Search & filtering
• Notifications
The architecture should be scalable so that additional modules can easily be integrated in future versions.

## 1.6 Vision

The long-term vision of TechFix is to become the primary digital facility management platform of Medeniyet Technopark.
The platform should provide a seamless experience for companies while giving administrators complete control over facility operations.
In future versions, TechFix can evolve into a comprehensive Smart Facility Management Platform by integrating preventive maintenance, AI-assisted issue analysis, equipment management, QR asset tracking and analytics.

## 2. System Analysis

### 2.1 System Purpose

TechFix is designed to provide a centralized, efficient and user-friendly platform for reporting and managing facility-related issues within Medeniyet Technopark.
The system connects companies and administrators through a single digital platform, eliminating the need for manual communication methods.
The primary purpose of the platform is to simplify issue reporting while enabling administrators to manage every report from one centralized dashboard.
The application is designed to be intuitive, fast and accessible for users with any level of technical experience.

### 2.2 Target Users

The system is intended to be used by two primary user groups.
Companies
Companies located inside Medeniyet Technopark will use the platform to report issues related to the facilities they occupy.
These users do not need an account.
Instead, they access the system by scanning a QR code placed on company doors or designated locations inside the Technopark.
Their interaction with the platform is intentionally simple and requires only a few steps.
Administrators
Administrators are responsible for reviewing, organizing, managing and resolving submitted reports.
They access a dedicated management dashboard protected by an administrator access code.
Only administrators have permission to manage reports and perform administrative actions.

### 2.3 User Permissions

Company User
Company users are allowed to:
• Scan the QR code
• Access the reporting form
• Select company/location
• Choose an issue category
• Write issue details
• Upload one or multiple photos
• Submit a report
• Receive a tracking code
• Check report status using the tracking code
Company users cannot:
• View reports submitted by other companies
• Edit reports after submission
• Access administrator functions

Administrator

Administrators are allowed to:
• Access the administrator dashboard
• View all submitted reports
• Search reports
• Filter reports
• Update report status
• Add administrator notes
• Assign priority levels
• Mark reports as resolved
• Archive completed reports
• View dashboard statistics
Administrators have full access to every report submitted through the system.

2.4 Authentication Model

The authentication model of TechFix is intentionally designed to minimize complexity for company users while maintaining administrative security.
Company Access
Companies are not required to create user accounts.
They do not need usernames or passwords.
Instead, company employees simply scan a QR code that opens the reporting interface.
This approach minimizes the time required to submit an issue and encourages users to report problems immediately.
Administrator Access
Administrator access is protected by a dedicated administrator access code.
Only authorized management personnel can enter the administration dashboard.
The administrator dashboard is completely separated from the public reporting interface.
Future versions of the platform may replace the shared administrator code with individual administrator accounts without changing the overall system architecture.

2.5 QR Code Workflow

The QR code is the main entry point of the system.
QR codes are placed on company doors and selected locations throughout Medeniyet Technopark.
When scanned, the QR code opens the TechFix reporting page.
The reporting process should require no more than one minute.
The QR code should always direct users to the latest version of the reporting interface.

## 2.6 System Workflow

### Company Workflow

Scan QR Code

↓

Open TechFix

↓

Select Company / Location

↓

Select Category

↓

Describe Issue

↓

Upload Photos (Optional)

↓

Submit Report

↓

Receive Tracking Code

### Administrator Workflow

Administrator Access

↓

Enter Administrator Code

↓

Dashboard

↓

View New Reports

↓

Review Issue

↓

Assign Priority

↓

Update Status

↓

Resolve Issue

↓

Archive Report

## 2.7 Report Lifecycle

Every report follows the same lifecycle.

New

↓

Under Review

↓

In Progress

↓

Resolved

↓

Archived

Every status change should be recorded by the system.

## 2.8 Functional Requirements

The system shall:

- Allow users to create reports without registration.
- Generate a unique tracking code for every report.
- Support image uploads.
- Allow administrators to manage reports.
- Store all reports permanently.
- Provide search and filtering functionality.
- Allow administrators to update report status.
- Support future scalability.

## 2.9 Non-Functional Requirements

The platform should:
• Be easy to use.
• Be responsive on desktop, tablet and mobile devices.
• Load quickly.
• Be intuitive for first-time users.
• Support future feature expansion.
• Maintain a clean and consistent user experience.
• Require minimal user interaction.

## 2.10 Project Success Criteria

The project will be considered successful if:
• Companies can submit reports in less than one minute.
• Administrators can quickly identify new reports.
• Every report is traceable.
• Issue management becomes centralized.
• The interface remains simple and intuitive.
• The platform improves communication efficiency inside Medeniyet Technopark.

# 3. System Features & Modules

## 3.1 Welcome Page

The Welcome Page is the first screen users see after opening TechFix.
This page should immediately communicate the purpose of the platform and guide users toward the correct action.
The interface must be clean, modern and easy to understand.
The page should contain only the essential actions.
Users should immediately understand that the system is designed for reporting facility-related issues inside Medeniyet Technopark.
The page should contain:
• TechFix Logo
• Project Title
• Short Description
• Create Report Button
• Administrator Access Button
• Report Tracking Button

## 3.2 QR-Based Access

The platform is designed around QR-code accessibility.
QR codes will be placed on company doors and selected locations throughout Medeniyet Technopark.
Users simply scan the QR code.
No application download is required.
No account creation is required.
No registration is required.
Scanning the QR code should immediately open the reporting page.
The reporting process should require less than one minute.

## 3.3 Report Creation Module

This is the most important module of the platform.
Users should be able to report an issue quickly with minimum effort.
The report form should include:
Company / Location
Issue Category
Issue Description
Photo Upload
Submit Button
The interface should encourage users to provide sufficient information while remaining simple.
The report form should not overwhelm users with unnecessary fields.

## 3.4 Issue Categories

The system should organize reports into categories.
Categories include:
Electrical
Internet / Network
Water & Plumbing
Air Conditioning
Elevator
Cleaning
Security
Common Areas
Equipment
Suggestion
Request
Other
Administrators should be able to filter reports using these categories.

### 3.5 Photo Upload

Users should be able to upload one or multiple photos.
Photos help administrators understand the issue before taking action.
Uploading photos should remain optional.
The system should automatically compress large images if necessary.
Supported formats:
JPG
PNG
WEBP

### 3.6 Report Submission

After clicking Submit:
The system should:
Validate required fields.
Upload images.
Generate a unique tracking code.
Store the report.
Notify administrators.
Display a success page.

### 3.7 Tracking Code

Each report receives a unique tracking code.
Example:
TFX-2026-000153
The tracking code allows users to check the progress of their report without creating an account.

### 3.8 Report Tracking Page

Users can enter their tracking code.
The system displays:
Current Status
Submission Date
Issue Category
Location
Administrator Notes
Completion Date
The tracking page should remain read-only.
Users cannot edit reports after submission.

### 3.9 Administrator Dashboard

The Administrator Dashboard is the central management panel.
Administrators should immediately see:
Total Reports
New Reports
Reports in Progress
Resolved Reports
Critical Reports
Recent Reports
Today's Statistics
Recent Activities
The dashboard should prioritize important information.

### 3.10 Report Management

Administrators should be able to:
Open reports
Review issue details
View uploaded photos
Change report status
Add internal notes
Assign priority
Archive reports
Delete invalid reports
Every administrator action should be recorded.

3.11 Search & Filtering

The dashboard should support advanced search.
Administrators should filter reports by:
Category
Status
Priority
Company
Location
Date
Keyword
Tracking Code

3.12 Report Priority

Administrators can assign one of four priority levels.
Low
Medium
High
Critical
Priority helps administrators organize maintenance work efficiently
.
3.13 Notifications

Administrators should receive notifications whenever:
A new report is submitted.
A critical report is created.
A report remains unresolved for too long.
Notifications should be visible inside the dashboard.
Future versions may include email or mobile push notifications.

3.14 Activity History

Every report should maintain a complete history.
History includes:
Creation Date
Status Changes
Administrator Notes
Resolution Date
Every action should remain permanently recorded.

3.15 Administrator Notes

Administrators should be able to add private notes.
These notes are only visible to administrators.
Example:
"Technician informed."
"Waiting for spare parts."
"Problem reproduced."

3.16 Report Resolution
When the issue has been solved:
Administrator marks the report as Resolved.
Optionally adds a resolution note.
The report moves to the archive.
Users checking their tracking code should immediately see the updated status
.
3.17 Statistics

The dashboard should display:
Number of reports
Most common issue category
Average resolution time
Reports by company
Monthly reports
Daily reports
Open vs Resolved reports
Critical issues

3.18 Archive

Resolved reports should remain archived.
Administrators should be able to:
Search archived reports.
Review previous issues.
Analyze recurring problems.
Export historical records in future versions.

3.19 Error Handling

The system should clearly inform users when:
Required fields are missing.
Images cannot be uploaded.
Tracking code is invalid.
Server connection fails.
Unexpected errors occur.
Every error message should be understandable.

### 3.20 Future Expansion

The platform architecture should support future modules without requiring major redesign.
Potential future modules include:

- AI Assistant
- Maintenance Scheduling
- Equipment Management
- QR Equipment Tracking
- Technician Management
- Document Management
- Meter Management
- Predictive Maintenance
- Mobile Application
- IoT Integration
- Advanced Analytics

### 3.21 Utility Meter Reading Module

The platform shall include a Utility Meter Reading module for recording electricity and natural gas meter readings.

Authorized personnel can periodically upload meter reading information by taking a photo of the meter.

Each record should include:

- Company / Location
- Meter Type (Electricity / Natural Gas)
- Meter Photo
- Optional Reading Value
- Notes (Optional)
- Upload Date
- Uploaded By (Future Version)

The uploaded records should be stored securely and remain accessible through the administrator dashboard.

Administrators should be able to review historical meter records, compare previous uploads and monitor utility consumption over time.

The system architecture should allow future integration with automatic meter reading technologies.

### 3.22 Administrator Meter Management

Administrators should have a dedicated section for managing uploaded meter records.

The dashboard should allow administrators to:

- View uploaded meter photos
- Filter by company
- Filter by meter type
- Filter by upload date
- Search historical records
- Download images
- View meter history

Future versions may support automatic consumption analysis.

# 4. Screen Specifications

## 4.1 Welcome Screen

Purpose
The Welcome Screen is the entry point of TechFix.
Its purpose is to immediately explain the system and guide users toward the correct action.
Users should instantly understand that TechFix is a platform for reporting facility-related issues inside Medeniyet Technopark.
The interface should contain minimal content while highlighting the two primary actions.
Components
TechFix Logo
Project Title
Short Description
Create Report Button
Administrator Access Button
Report Tracking Button
Footer
User Actions
Users can
• Create a new report
• Enter the administrator panel
• Track an existing report

## 4.2 Create Report Screen

Purpose
Allow users to report issues as quickly as possible.
The page should minimize typing and reduce unnecessary interactions.
Components
Company / Location Selector
Issue Category Selector
Issue Description
Photo Upload
Submit Button
Cancel Button

User Actions
Users can
Choose company
Choose category
Write description
Upload photos
Submit report

## 4.3 Success Screen

Purpose
Confirm that the report has been successfully submitted.
The page should reassure the user that the issue has been recorded.
Components
Success Icon
Confirmation Message
Tracking Code
Track Report Button
Back to Home Button
User Actions
Copy tracking code
Track report
Return home

## 4.4 Report Tracking Screen

Purpose
Allow users to check the progress of previously submitted reports.
No login should be required.
Components
Tracking Code Input
Search Button
Report Details
Current Status
Administrator Notes
Submission Date
Completion Date
User Actions
Enter tracking code
View report status

## 4.5 Administrator Login Screen

Purpose
Provide secure access to the administrator dashboard.
Only authorized personnel should access this page.
Components
Administrator Code Input
Login Button
Back Button
User Actions
Enter administrator code
Login
Return to home

## 4.6 Administrator Dashboard

Purpose

Provide administrators with a complete overview of the entire platform.

This page should become the central workspace.

Components

Sidebar Navigation

Dashboard Statistics

Recent Reports

Priority Reports

Recent Activities

Quick Actions

Notifications

Search Bar

Dashboard Cards

Total Reports

Open Reports

Reports In Progress

Resolved Reports

Critical Reports

Today's Reports

User Actions

Open reports

Search

Filter

Manage reports

Navigate to modules

## 4.7 Report Detail Page

Purpose

Display every detail related to a report.

Administrators should perform all report management from this page.

Components

Tracking Code

Company

Location

Category

Description

Uploaded Images

Status

Priority

Administrator Notes

Timeline

Action Buttons

User Actions

Change status

Assign priority

Add notes

Archive report

Delete report

## 4.8 Reports Page

Purpose
Display every submitted report.
Components
Search
Filters
Data Table
Pagination
Export Button
Filters
Status
Category
Company
Priority
Date
Keyword

## 4.9 Statistics Page

Purpose
Provide detailed analytics regarding facility issues.
Components
Charts
Issue Categories
Monthly Reports
Average Resolution Time
Open vs Closed Reports
Top Companies

## 4.10 Notifications

Purpose
Display administrator notifications.
Components
Notification List
Priority Badge
Date
Open Report Button

## 4.11 Settings

Purpose
Allow administrators to configure system preferences.
Components
Administrator Code
Categories
Status Types
Notification Settings
General Settings

4.12 Error Pages

The application should contain
404 Page
500 Page
Offline Page
Permission Denied Page
Each page should remain consistent with the overall design language
.
4.13 Responsive Behaviour

Every screen should adapt smoothly to
Desktop
Tablet
Mobile
No functionality should be lost on smaller devices.

4.13 Utility Meter Upload Screen
Purpose

Allow authorized personnel to upload electricity and natural gas meter photos quickly.

Components

Company Selector

Meter Type

Photo Upload

Reading Value (Optional)

Notes

Submit Button

## 4.14 Navigation

Navigation should remain simple.
Users should never feel lost.
Important actions should always be visible.
Back navigation should always be available.

## 4.15 Screen Consistency

Every page should follow the same visual language.
Components should be reused throughout the application.
Buttons, cards, forms and dialogs should remain consistent across the system.

## 4.16 Performance Expectations

The interface should feel fast.
Users should receive immediate visual feedback after every action.
Loading indicators should be displayed whenever necessary.
The application should never leave the user uncertain about the current system state.

## 4.18 Meter Management Dashboard

Purpose

Allow administrators to monitor and manage uploaded utility meter records.

Components

- Meter Records List
- Photo Preview
- Company Filter
- Meter Type Filter
- Upload Date Filter
- Search
- Record Details
- Download Image

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## 5. User Experience & Design Vision

### 5.1 Design Philosophy

,
TechFix should not feel like a traditional government or enterprise management system.
Instead, it should feel like a modern, premium and professional software platform designed for everyday use.
The application should create confidence from the first screen while remaining simple enough for users of all technical backgrounds.
Every interaction should feel smooth, natural and intuitive.
The design should encourage users to complete tasks quickly without confusion.

### 5.2 Overall User Experience

The platform is designed around one principle:
Simplicity first.
Users should never need training before using the application.
Every screen should clearly communicate its purpose.
The reporting process should be fast, effortless and require the minimum possible number of steps.
Users should never feel overwhelmed by unnecessary information.

### 5.3 First Impression

When opening TechFix for the first time, users should immediately understand:
- What the platform is
- Why it exists
- What they can do
- How to start

The interface should communicate professionalism, trust and clarity.
The first impression should encourage users to continue using the platform.

### 5.4 Reporting Experience

Creating a report should take less than one minute.
The interface should minimize typing.
Only essential information should be requested.
Users should always know what to do next.
Uploading a photo should be simple.
Submitting a report should provide immediate confirmation.

### 5.5 Administrator Experience

Administrators spend significantly more time inside the platform than regular users.
For this reason, the dashboard should prioritize efficiency over decoration.
Important information should always be visible.
Searching, filtering and managing reports should require as few clicks as possible.
Administrators should never need to navigate through multiple unnecessary pages to complete routine tasks.

## 5.6 Dashboard Philosophy

The dashboard should function as a command center.
Administrators should immediately understand the current state of the facility.
Critical information should always be visible first.
The dashboard should highlight:
New reports
Critical reports
Pending issues
Recent activity
Statistics
The interface should support quick decision making.

## 5.7 Navigation

Navigation should always remain predictable.
Users should never wonder where they are.
Moving between pages should require minimal effort.
Frequently used actions should always be easily accessible.

## 5.8 Forms

Forms should be clean and straightforward.
Every field should have a clear purpose.
Avoid unnecessary questions.
Labels should be understandable.
Validation messages should help users rather than confuse them.

## 5.9 Feedback

Every user action should receive immediate feedback.
Examples include:
Report submitted
Photo uploaded
Status updated
Error occurred
Administrator note saved
The system should never leave users uncertain about whether an action was successful.

## 5.10 Error Handling

Errors should be explained in plain language.
The interface should always provide guidance on how to solve the problem.
Users should never encounter technical error messages.
The application should remain calm and reassuring even when something goes wrong.

## 5.11 Accessibility

The platform should be usable by users with different levels of technical knowledge.
Buttons should be easy to identify.
Text should remain readable.
Interactive elements should provide clear feedback.
The platform should prioritize usability over visual complexity.

## 5.12 Consistency

Every page should follow the same interaction patterns.
Buttons should behave consistently.
Forms should remain familiar.
Navigation should remain predictable.
Users should never need to relearn the interface when moving between screens.

## 5.13 Performance Perception

The application should always feel responsive.
Loading indicators should appear whenever necessary.
Transitions should be smooth but subtle.
Animations should support usability rather than distract the user.

## 5.14 Emotional Experience

Using TechFix should create feelings of:
Confidence
Professionalism
Reliability
Efficiency
Clarity
Users should feel that the platform is trustworthy and well designed.

## 5.15 Design Inspiration

The application should reflect the quality of modern commercial software.
It should avoid looking like an academic project or an outdated enterprise system.
The interface should appear polished, professional and ready for real-world deployment.

## 5.16 Design Principles

Throughout the application, the following principles should always be respected:
Simplicity over complexity.
Clarity over decoration.
Consistency over variation.
Usability over visual effects.
Efficiency over unnecessary interactions.
Professionalism over excessive styling.
Scalability over short-term solutions.
Every design decision should support these principles.

## 5.17 Mobile Experience

Although the primary usage is expected on desktop devices, the platform must provide a seamless experience on tablets and smartphones.
Users should be able to create reports comfortably from any mobile device immediately after scanning the QR code.
The mobile interface should maintain the same quality and usability as the desktop version.

## 5.18 Final Design Goal

The final product should feel like a premium software platform developed by an experienced software company.
Every screen should demonstrate attention to detail, consistency and high usability.
The application should be suitable for deployment in a real organizational environment without requiring major design changes.

# 6. AI Development Instructions

## 6.1 Purpose

This document serves as the primary reference for designing and developing the TechFix platform.
All design and development decisions should follow the requirements described in this document.
If any functionality is unclear, preserve the overall product vision and prioritize simplicity, usability and consistency

## 6.2 Development Objective

The goal is to build a production-ready web application rather than a prototype or an academic project.
The final product should be suitable for deployment in a real organizational environment.
The application should demonstrate professional software engineering practices and be easy to maintain, extend and improve.

## 6.3 General Development Principles

The application should be:
Modular
Scalable
Maintainable
Reusable
Responsive
Secure
Easy to use
Consistent
Every new feature should integrate naturally into the existing architecture without requiring major redesign.

## 6.4 User-Centered Design

Every development decision should prioritize the user experience.
Users should never perform unnecessary actions.
Every feature should solve a real problem.
Interfaces should reduce complexity rather than increase it.

## 6.5 Design Rules

The interface should always remain:
Clean
Minimal
Modern
Professional
Balanced
Readable
Well organized
Avoid:
Cluttered layouts
Outdated enterprise interfaces
Unnecessary popups
Overly decorative elements
Confusing navigation
Inconsistent pages

## 6.6 Interface Consistency

Every page should follow the same visual language.
Navigation should always behave consistently.
Buttons should maintain the same behavior.
Forms should remain familiar.
Dialog windows should follow identical interaction patterns.
Users should never need to learn different interfaces for different modules.

## 6.7 Simplicity

Whenever multiple implementation options exist, choose the simplest solution that provides the best user experience.
Avoid unnecessary complexity.
The platform should feel effortless.

## 6.8 Scalability

Every module should be developed with future expansion in mind.
Future additions such as:
AI Assistant
Equipment Management
Preventive Maintenance
IoT Integration
Analytics
Mobile Application
should be possible without redesigning the entire system.

## 6.9 Administrator Experience

Administrators are the primary long-term users of the system.
Their workflow should be optimized for speed.
Frequently used actions should require the fewest possible interactions.
Information should always be accessible within a minimal number of clicks.

## 6.10 Company Experience

Company users should only focus on reporting issues.
No technical knowledge should be required.
The reporting process should remain fast and intuitive.
The interface should minimize typing whenever possible.

## 6.11 Error Prevention

The system should prevent user mistakes whenever possible.
Provide validation before submission.
Explain problems clearly.
Avoid technical language.
Help users recover from errors.

## 6.12 Performance

The application should feel fast.
Loading times should remain minimal.
The interface should provide immediate feedback after every interaction.

## 6.13 Accessibility

The application should remain usable by people with varying technical abilities.
Interactive elements should be easy to identify.
Navigation should remain predictable.
Forms should remain easy to complete.

## 6.14 Security

Administrator access must remain separated from the public reporting interface.
Administrative actions should require authentication.
Sensitive operations should only be available to authorized administrators.
Future versions should support stronger authentication methods without redesigning the application
.

## 6.15 Maintainability

The project should remain easy to maintain.
Future developers should understand the project structure without extensive documentation.
Features should remain independent whenever possible.

6.16 Long-Term Vision

TechFix should evolve into a complete Smart Facility Management Platform capable of managing every operational process inside Medeniyet Technopark.
Future development should always respect the original vision while remaining flexible enough to support new technologies and organizational needs.

6.17 Final Instructions for AI

When generating designs or application structures, always prioritize:
Simplicity
Usability
Consistency
Scalability
Professionalism
Accessibility
Maintainability
Do not introduce unnecessary complexity.
Do not sacrifice usability for visual effects.
Every design and development decision should support the overall mission of TechFix: enabling fast, simple and efficient facility issue management for Medeniyet Technopark.

# 7. System Architecture & Business Logic

## 7.1 System Overview

TechFix is designed as a centralized issue management platform.
The system connects companies operating inside Medeniyet Technopark with the facility management team through a simple and efficient reporting workflow.
Every issue reported through the platform follows a structured lifecycle, from submission to resolution, ensuring transparency and traceability throughout the process.
The system should be modular, scalable and capable of supporting future expansion without requiring major architectural changes.

## 7.2 Core Entities

The platform is built around the following core entities:
Report
Represents a submitted issue or request.
Each report should contain:
Unique Tracking Code
Company
Location
Category
Description
Attached Photos
Current Status
Priority
Submission Date
Last Updated Date
Administrator Notes
Resolution Date
Company
Represents a company located inside Medeniyet Technopark.
Each company should contain:
Company Name
Building
Floor
Office Number
QR Code Reference

Administrator

Represents authorized personnel responsible for managing reports.
Administrator information includes:
Name
Role
Access Code (or authentication method)
Activity History

Category

Each report belongs to one predefined category.
Examples:
Electrical
Internet / Network
Plumbing
Air Conditioning
Elevator
Cleaning
Security
Equipment
Suggestion
Other

Attachment

Represents uploaded files.
Supported attachment types:
Images
Documents (future versions)
Each attachment belongs to one report.

7.3 Report Lifecycle
Every submitted report follows the same business process.

QR Scan
↓
Create Report
↓
Validation
↓
Report Created
↓
Tracking Code Generated
↓
Administrator Notification
↓
Administrator Review
↓
Status Updated
↓
Issue Resolved
↓
Archived

Every transition should be recorded in the report history.

## 7.4 Business Rules

The following business rules must always apply.
Rule 1
Every report must receive a unique tracking code.
Rule 2
A report cannot exist without a selected category.
Rule 3
A report cannot be submitted without a description.
Rule 4
Photo upload is optional but recommended.
Rule 5
Only administrators can change report status.
Rule 6
Users cannot edit reports after submission.
Rule 7
Archived reports remain searchable.
Rule 8
Every administrator action should be logged.
Rule 9
The system should prevent duplicate report submissions whenever possible.
If a very similar issue already exists for the same company and category within a recent time period, the user should be warned before creating another report.

## 7.5 Notification Logic

Whenever a new report is submitted:
The system should:
Save the report
Generate tracking code
Notify administrators
Display confirmation to the user
Whenever an administrator updates a report:
The system should:
Save the update
Record the action in history
Update report status
Future versions may notify the reporting company when the status changes.

7.6 Report Status Logic

Available statuses:
New
↓
Under Review
↓
Assigned
↓
In Progress
↓
Resolved
↓
Archived
The administrator may only move reports through valid status transitions
.
7.7 Administrator Workflow

Administrator opens dashboard.
↓
Views new reports.
↓
Reads issue details.
↓
Reviews uploaded images.
↓
Determines priority.
↓
Updates status.
↓
Adds administrator notes.
↓
Marks report as resolved.
↓
Archives report.

## 7.8 Company Workflow

Company employee scans QR code.
↓
TechFix opens.
↓
Company/location is selected.
↓
Category is selected.
↓
Issue description is entered.
↓
Photos are uploaded (optional).
↓
Report is submitted.
↓
Tracking code is displayed.

## 7.9 Tracking Logic

Users can enter their tracking code at any time.
The system should display:
Current Status
Submission Date
Category
Company
Resolution Notes (if available)
Completion Date (if resolved)
Users cannot modify reports from the tracking page.

## 7.10 Data Integrity

The system should always ensure:
Every report belongs to exactly one company.
Every report belongs to exactly one category.
Every attachment belongs to one report.
Every administrator action is recorded.
Every tracking code is unique.
Reports cannot exist without required information.

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### 7.11 Future Scalability

The architecture should support future modules, including:
Technician Management
Preventive Maintenance
Equipment Inventory
Asset QR Codes
AI Assistant
Predictive Maintenance
Mobile Application
IoT Sensors
Smart Notifications
Analytics Dashboard
Multi-Technopark Support
Multi-language Support
These modules should integrate seamlessly without redesigning the existing system

### 7.12 System Vision

TechFix is not intended to be only an issue reporting website.
The long-term vision is to become a complete Smart Facility Management Platform capable of managing maintenance operations, equipment tracking, facility communication and operational workflows within Medeniyet Technopark through a centralized, intelligent and scalable system.

## 8. Functional Requirements

The system shall:

- Allow users to submit reports without registration.

- Allow users to access the platform by scanning a QR code.

- Generate a unique tracking code for every submitted report.

- Allow users to upload one or more images.

- Allow administrators to securely access the management dashboard.

- Allow administrators to review, filter and manage all reports.

- Allow administrators to update report status.

- Store all reports permanently.

- Maintain a complete activity history for every report.
- Prevent duplicate report submissions whenever possible.
- Allow users to track their reports using the tracking code.
- Support future system expansion without redesigning the existing architecture.

## 9. Non-Functional Requirements

The application should:

- Be responsive across desktop, tablet and mobile devices.
- Provide a fast and intuitive user experience.
- Load quickly with minimal waiting time.
- Be scalable for future feature additions.
- Maintain a consistent interface throughout the platform.
- Ensure data integrity and reliability.
- Be easy to maintain and extend.
- Provide high usability for users with different technical backgrounds.

## 10. Assumptions & Constraints

### Assumptions

- QR codes will be placed by Medeniyet Technopark.
- Companies will scan the QR code to access the reporting platform.
- Administrator access codes will only be distributed to authorized personnel.
- Internet access will be available during report submission.
- Users will provide accurate report information.

Constraints

• Version 1 focuses only on issue reporting and management.

• Company user accounts are intentionally excluded.

• Individual administrator accounts are reserved for future versions.

• Technician management is outside the scope of Version 1.

• Equipment inventory management is outside the scope of Version 1.

11. Success Criteria

The project will be considered successful if:

• Users can submit a report in less than one minute.

• Administrators can efficiently manage reports.

• Every report remains traceable.

• Communication becomes centralized.

• Issue resolution becomes faster.

• The platform is intuitive for first-time users.

• The application is ready for deployment within Medeniyet Technopark

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12. Out of Scope (Version 1)

The following features are intentionally excluded from the first version:

• User registration

• Individual company accounts

• Technician mobile application

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• Email notifications
• SMS notifications
• AI automatic responses
• Equipment inventory management
• IoT sensor integration
• Payment or billing modules
• Advanced predictive analytics

These features may be considered in future versions

13. Complete User Flow

COMPANY USER

Scan QR Code
↓
Open TechFix
↓
Select Company / Location
↓
Select Issue Category
↓
Write Description
↓
Upload Photos (Optional)
↓
Submit Report
↓
Receive Tracking Code
↓
Track Report Status

-----------------------------------------------------

ADMINISTRATOR

Administrator Access
↓
Enter Administrator Access Code
↓
Dashboard
↓
View Reports
↓
Review Report
↓
Assign Priority
↓
Update Status
↓
Resolve Report
↓
Archive Report

14. Final Statement

This document serves as the single source of truth for the TechFix project.
All design, development and implementation decisions must strictly follow the requirements defined in this document.
The primary objective is to deliver a modern, scalable, user-friendly and production-ready Smart Facility Issue Management Platform for Medeniyet Technopark.
Any future improvements should remain fully compatible with the project's architecture, business logic and long-term vision.

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